Imbalance-Aware Training and Wavelength Occlusion for Grapevine Spectral Classification

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***Abstract*—Class imbalance can conceal failures in spectral classifiers when evaluation emphasizes overall accuracy. This paper investigates imbalance-aware training for a six-class grapevine symptom classification task using an existing one-dimensional residual network. The available dataset contains 180 image-level spectral signatures, each represented by 169 reflectance features and 169 derivative features. A fixed split assigns 108 samples to training and 36 each to validation and testing. Three configurations retain the same weighted cross-entropy objective and architecture: ordinary shuffled training, inverse-frequency sampling, and inverse-frequency sampling with conservative spectral perturbations. The perturbations combine Gaussian noise, multiplicative intensity scaling, and additive baseline shifts while respecting the different roles of reflectance and derivative features. Checkpoints are selected using validation macro-averaged F1 (macro-F1). In one seeded, 60-epoch run per configuration, test macro-F1 increases from 0.3353 to 0.4111 with sampling and to 0.4177 with sampling and augmentation. The augmented configuration retains baseline accuracy of 66.67%. Both sampling configurations correctly classify one of three buffalo treehopper samples, but neither correctly identifies water stress or wood diseases. Paired wavelength-region occlusion further exposes the sensitivity of an incorrect prediction. These findings support class-resolved evaluation and reproducible experimentation, while the small test set, single seed, and legacy label-derived loss weights limit claims of generalization.**

***Index Terms*—Hyperspectral data, grapevine symptom classification, class imbalance, residual networks, spectral augmentation, occlusion sensitivity.**

I. Introduction

Spectral measurements provide a basis for investigating plant symptoms beyond visible appearance. Public grapevine hyperspectral resources include healthy leaves and several biotic and abiotic symptom categories, enabling controlled studies of spectral classification [1]. Distinguishing confounding symptoms remains important: recent work examines grapevine yellows and related stresses using leaf spectra, while accounting for differences between grapevine cultivars [2]. Classification of an acquisition's symptom label, however, is not equivalent to confirming a pathogen, detecting presymptomatic infection, or validating a field diagnostic device.

An additional challenge is the unequal availability of labeled examples. In the dataset examined here, flavescence dorée and healthy samples account for 120 of 180 observations. A classifier can therefore achieve a seemingly useful accuracy while never recognizing several less frequent classes. Research on imbalanced convolutional networks distinguishes resampling from cost-sensitive optimization and cautions against evaluating imbalance solely through aggregate correctness [3]. Whether these measures benefit a particular small spectral dataset must be established empirically.

This work examines the Spec-Crop implementation using a fixed one-dimensional residual network, following the residual-learning principle of He *et al.* [4]. The scope is a reproducible comparison of training and evaluation strategies, rather than a new network architecture. The study asks whether inverse-frequency sampling changes class-resolved performance when a weighted-loss baseline is retained, and whether small spectral perturbations provide an additional benefit.

The contributions are threefold. First, the training framework integrates training-derived sampling probabilities and conservative perturbations of paired reflectance and derivative features without modifying the existing data partitions or architecture. Second, a matched three-configuration experiment selects checkpoints by validation macro-averaged F1 (macro-F1) and reports both aggregate and class-specific outcomes. Third, wavelength-region occlusion and automated reproducibility checks make prediction sensitivity and implementation behavior inspectable. The measured improvements are reported together with the persistent failures on water stress and wood diseases.

II. Related Work

A. Spectral Classification of Grapevine Symptoms

The resource described by Ryckewaert *et al.* [1] contains laboratory hyperspectral acquisitions covering multiple cultivars and symptom categories. It supports investigation of confounding factors but explicitly presents small sample availability and unequal category counts as limitations. Alisaac *et al.* [2] subsequently studied discrimination of grapevine yellows and symptomatically similar stresses using mean leaf spectra, with cultivar-dependent results. These studies motivate classification beyond a healthy-versus-diseased distinction. Their acquisition protocols, labels, and evaluation designs differ from the present six-class split; their reported scores are therefore not numerical baselines for this experiment.

B. Rebalancing and Spectral Perturbations

Resampling changes the distribution of examples presented to an optimizer, whereas loss weighting changes their contribution to the objective. Buda *et al.* [3] compared several imbalance strategies for convolutional models, and Cui *et al.* [5] proposed weighting based on the effective number of samples. The present implementation uses simple inverse-frequency sampling and the existing class weights; it does not implement effective-number weighting. Its comparison specifically measures the addition of sampling to an already weighted loss.

Spectral augmentation requires attention to signal structure. Bjerrum, Glahder, and Skov [6] investigated augmentation for convolutional chemometric models in a near-infrared regression setting. That work motivates physically interpretable perturbations, but does not establish their effectiveness for grapevine classification. Here, multiplicative scaling and constant offsets are applied in reconstructed raw units, and derivative features receive corresponding perturbations. Interpolation between different training spectra is not used. The transformation amplitudes are conservative design choices rather than parameters calibrated from repeated sensor acquisitions.

C. Prediction Sensitivity

Occlusion analysis examines how removing a localized input region changes a model output; Zeiler and Fergus [7] used such perturbations to study convolutional-network behavior. The present utility applies this principle to consecutive wavelengths and masks both representations of each wavelength together. It measures dependence of a fixed predicted-class score on the replacement intervention. It does not establish a causal relationship between a wavelength region and a disease, particularly when the original prediction is incorrect.

III. Data and Problem Formulation

A. Data Provenance and Representation

The preprocessing code identifies the public dataset with DOI 10.57745/WW7TY7 [8], associated with [1]. The local processed subset contains 180 acquisitions and six symptom labels (Table I). The original acquisition study used a Specim IQ camera under controlled conditions; this is provenance of the source data, not hardware built or evaluated in the present project.

The recorded preprocessing procedure loads reflectance cubes, retains wavelengths between 400 and 900 nm, and averages over both spatial dimensions. No explicit leaf segmentation mask is applied by this code. It then computes smoothed reflectance and first-derivative features using a Savitzky–Golay filter [9], with window length seven and polynomial degree two. Smoothing and differentiation are computed separately from the image-mean spectrum. Derivatives use unit sample spacing, so they are not calibrated derivatives per nanometer. The supplied CSVs contain 169 paired wavelength positions, whose integer-valued column labels span 400–899 nm.

TABLE I
Class Distribution in the Fixed Data Partitions

Class	Train	Val.	Test	Total
Buffalo treehopper	10	4	3	17
Flavescence dorée	48	16	16	80
Green leafhopper	10	3	3	16
Healthy	24	8	8	40
Water stress	8	2	3	13
Wood diseases	8	3	3	14
Total	108	36	36	180

Each input is a concatenated vector

$$\mathbf{x} = [\mathbf{r}^\mathsf{T}, \mathbf{d}^\mathsf{T}]^\mathsf{T} \in \mathbb{R}^{338}, \tag{1}$$

where $\mathbf{r}$ and $\mathbf{d}$ contain 169 reflectance and derivative features, respectively. Additional spectral indices in the repository are not model inputs. Labels $y \in \{0, \ldots, 5\}$ follow the order in Table I. The classifier produces logits $\mathbf{z}_\theta(\mathbf{x})$, probabilities $p_\theta(c \mid \mathbf{x})$ through softmax, and prediction $\hat{y} = \arg\max_c p_\theta(c \mid \mathbf{x})$.

B. Fixed Partitions and Standardization

The original procedure uses stratified splitting with random state 42: 20% of the observations form the test set, followed by a 25% validation allocation from the remaining 80%. This yields a 60/20/20 split. Featurewise standardization uses means μ_j and scales s_j fitted on training data only:

$$x_j^{\mathrm{norm}} = (x_j - \mu_j)/s_j. \tag{2}$$

The saved scaler records 108 fitted observations. An audit reconstructed the documented split indices and transformed the local unnormalized feature table using the existing scaler, without refitting it. The resulting labels matched all three CSVs, and maximum absolute feature differences were below 2×10^{-15}. No nonfinite values, exact within-split duplicate rows, or exact cross-split row overlaps were found. These checks establish artifact consistency, not biological independence between acquisitions.

IV. Methodology and Implementation

A. System Workflow and Residual Classifier

Fig. 1 separates fixed data preparation from training and final evaluation. Only the training loader applies resampling or augmentation. Validation determines the selected epoch; test predictions are generated afterward from the saved checkpoint. CSV files store input data and histories, JSON files store configurations and metrics, and PyTorch state dictionaries store model parameters. The evaluated path is a local Python workflow; no database, web service, or deployed inference endpoint is required.

The network treats the complete 338-feature concatenation as one input channel. A convolutional stem is followed by four residual blocks, adaptive average pooling, dropout with probability 0.3, and a six-output linear layer. Each block contains two convolutions and batch normalization, with a

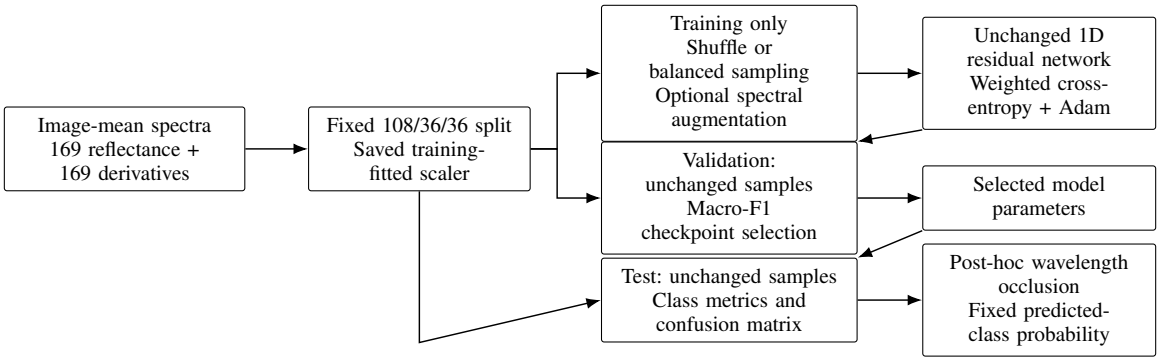


Fig. 1. Implemented experimental workflow. The split and scaler are reused without fitting on validation or test data. The separate diffraction-image extraction prototype is not part of this evaluated path.

TABLE II
UNCHANGED ONE-DIMENSIONAL RESIDUAL NETWORK

Stage	Channels	Length	Kernel / stride
Input	1	338	—
Stem	64	338	7 / 1
Residual block 1	64	338	7 / 1
Residual block 2	128	169	5 / 2
Residual block 3	256	85	3 / 2
Residual block 4	256	85	3 / 1
Average pool	256	1	Global
Dropout + linear	6 outputs	—	$p = 0.3$

Block stride applies to its first convolution and shortcut.

rectified linear unit after the first normalization and after residual addition. Channel or stride changes use a 1×1 convolution and normalization in the shortcut. Table II specifies the dimensions; the architecture contains 915,014 trainable parameters. It is unchanged across all experiments. Because the feature groups are concatenated, the convolution operates across the boundary between the last reflectance and first derivative feature; they are not separate input channels.

B. Weighted Loss and Class-Balanced Sampling

All compared configurations minimize weighted cross-entropy. For a minibatch $\mathcal{B}$, the implemented mean reduction is

$$\mathcal{L}_{\mathcal{B}}(\theta) = -\frac{\sum_{i\in\mathcal{B}} a_{y_i} \log p_\theta(y_i \mid \mathbf{x}_i)}{\sum_{i\in\mathcal{B}} a_{y_i}}, \tag{3}$$

where a_c is the retained weight for class c. In Table I order, the weights are approximately $(1.7647, 0.3750, 1.8750, 0.7500, 2.3077, 2.1429)$. The original code obtained them as $180/(6N_c)$, where N_c counts the full processed dataset. Consequently, this legacy objective incorporates held-out class-frequency information. It is preserved identically across configurations for comparability and is not presented as a strictly training-only loss design. A training-derived loss-weight option is implemented but is not the basis of the reported 60-epoch comparison.

Let n_c count training examples of class c. For balanced sampling, observation i receives unnormalized weight $w_i = 1/n_{y_i}$. Its sampling probability is

$$q_i = \frac{w_i}{\sum_{j=1}^{108} w_j} = \frac{1}{6n_{y_i}}. \tag{4}$$

Under (4), each class has probability $1/6$ per draw. PyTorch's `WeightedRandomSampler` draws 108 observations per epoch with replacement; individual epochs are balanced only in expectation. The baseline instead shuffles all 108 observations without replacement. Combining sampling with the retained weighted objective changes both exposure and loss contributions. It can overemphasize minority examples, so improved recall need not coincide with improved overall accuracy.

C. Conservative Spectral Augmentation

Training samples are returned to raw feature units using the inverse of (2). Let σ_{ref} denote the median, over reflectance bands, of the population standard deviation computed from raw training features. For the saved experiment, $\sigma_{\mathrm{ref}} = 0.0092146$. For each sampled spectrum, transformations are

$$\widetilde{\mathbf{r}} = a\mathbf{r} + \boldsymbol{\epsilon} + b\mathbf{1}, \tag{5}$$

$$\widetilde{\mathbf{d}} = a\mathbf{d} + D_{7,2}\boldsymbol{\epsilon}, \tag{6}$$

where $a \sim \mathcal{U}(0.98, 1.02)$, $b \sim \mathcal{U}(-0.005\sigma_{\mathrm{ref}}, 0.005\sigma_{\mathrm{ref}})$, and independent $\epsilon_j \sim \mathcal{N}(0, (0.005\sigma_{\mathrm{ref}})^2)$. The operator $D_{7,2}$ is the same seven-position, degree-two first-derivative filter used by preprocessing. A constant baseline shift is not added to derivative features. The transformed vector is standardized with the unchanged saved statistics. These operations approximate small intensity and baseline variation; they do not constitute a calibrated physical sensor model or a regenerated hyperspectral acquisition.

An optional mask replaces three consecutive wavelength positions in both feature groups with their training means, equivalent to zero in standardized space. Its application probability and width are configurable. This mask is disabled in the reported three-way experiment and exercised only in implementation tests. No validation or test sample is augmented

Algorithm 1 Fixed-split imbalance experiment

Require: Fixed train/validation/test data; saved scaler and class weights; mode m; seed s; epochs E
Ensure: Selected parameters θ^* and test metrics
Seed random generators; initialize the residual network
Count training labels and construct the mode-specific loader
Set $F^* \leftarrow -1$
for $e = 1, \ldots, E$ **do**
  for minibatch $\mathcal{B}$ from the training loader **do**
    if m enables augmentation **then**
      Apply (5)–(6) using saved scaling
    end if
    Update θ with Adam using (3)
  end for
  Evaluate unchanged validation data in inference mode
  if $F^{\mathrm{val}}_{1,\mathrm{macro}} > F^*$ **then**
    Save $\theta^* \leftarrow \theta$ and update F^*
  end if
  Record history and advance the learning-rate schedule
end for
Load θ^*; evaluate unchanged test data once
Save metrics, confusion matrix, configuration, and plots

during predictive evaluation. Post-hoc occlusion, described next, is a separate diagnostic intervention.

D. Checkpoint Selection and Occlusion

For class c, let TP_c, FP_c, and FN_c denote true positives, false positives, and false negatives. Precision P_c and recall R_c are their usual classwise ratios, with zero assigned when a relevant denominator is zero. The checkpoint criterion is

$$F_{1,\mathrm{macro}} = \frac{1}{6}\sum_{c=0}^{5} \frac{2\mathrm{TP}_c}{2\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}. \tag{7}$$

The criterion in (7) gives equal weight to all classes and penalizes both false alarms and missed cases. A checkpoint is saved only when validation macro-F1 strictly increases; ties retain the earliest epoch. Algorithm 1 summarizes the procedure.

For occlusion, the original prediction $\hat{c}$ is held fixed. A region W comprises five consecutive wavelength positions and their matching reflectance and derivative features. The utility replaces those standardized features with zero and computes

$$\Delta_W = p_{\theta^*}(\hat{c} \mid \mathbf{x}) - p_{\theta^*}(\hat{c} \mid M_W(\mathbf{x})), \tag{8}$$

In (8), M_W denotes training-mean replacement. A positive Δ_W means the intervention reduces the original predicted-class probability. Windows advance by two positions, and an endpoint window ensures coverage of the final wavelength. The 169 positions produce 83 windows in the reported example. Column names provide wavelength labels; integer labels approximate the original band centers.

V. Experimental Setup

A. Configurations and Reproducibility

Three configurations are compared: B, shuffled weighted-loss training; S, the same loss with balanced sampling; and A, balanced sampling with the perturbations in (5)–(6). Each is trained from scratch for 60 epochs with seed 42, batch size 16, and identical initialization settings. Adam [10] uses learning rate 10^{-3}, default moment coefficients $(0.9, 0.999)$, $\epsilon = 10^{-8}$, and no weight decay. The learning rate halves after epochs 20 and 40. No early stopping, pretrained weights, or architecture search is used. Each epoch contains seven minibatches, including a final batch of 12 observations.

The runs use Windows 11 and CPU execution with two PyTorch threads and zero loader workers. The recorded software versions are Python 3.12.14, PyTorch 2.14.0+cpu, NumPy 2.5.3, pandas 3.0.6, scikit-learn 1.6.1, and SciPy 1.18.1; Matplotlib 3.11.2 produces plots. Python, NumPy, and PyTorch generators and loader/sampler generators are seeded, and deterministic algorithms are required. Exact processor branding and memory capacity were not recorded with the training artifacts; no runtime, throughput, or hardware-efficiency claim is made.

The original partitions and normalization are reused without modification. SHA-256 fingerprints of the input splits accompany each configuration. The saved histories and checkpoints permit independent verification of selected epochs. Seven automated tests cover sampling, perturbation behavior, metrics, and occlusion, including two repeated two-epoch augmented runs with masking enabled and training-derived loss weights. Those repetitions produce identical histories, test metrics, and checkpoint tensors. They verify implementation reproducibility, not variability across independent training seeds.

B. Evaluation Protocol

Validation macro-F1 selects one checkpoint per configuration. All reported test scores refer to those selected checkpoints, rather than the final epoch. Accuracy counts correct predictions over all 36 test observations. Balanced accuracy is the arithmetic mean of class recall; all six classes occur in this test split, so it equals macro recall. Macro precision and macro-F1, per-class precision/recall/F1, prediction counts, and a fixed-order 6×6 confusion matrix are also recorded. Unpredicted classes are retained with zero precision and F1. Training history describes predictions made during optimization on the shuffled or sampled stream, potentially with augmentation; it is not a separate evaluation of the clean training set.

The comparisons use one seed and one fixed partition. No cross-seed standard deviations, confidence intervals, or significance tests are inferred from the deterministic repetitions. Configuration differences are interpreted as observed outcomes, not estimates of an expected population-level improvement. The older repository checkpoint is excluded from the matched comparison because its seed and selection conditions are not established by the saved run configurations.

VI. Results and Discussion

A. Aggregate Performance and Trade-offs

Table III gives the complete aggregate comparison. Balanced sampling raises test macro-F1 from 0.3353 to 0.4111, an absolute increase of 0.0758, while decreasing accuracy from 24/36 to 22/36 correct predictions. Adding augmentation yields macro-F1 0.4177 and restores 24/36 correct predictions.

Relative to the baseline, the latter increases balanced accuracy from 0.3785 to 0.4583. The augmentation-versus-sampling macro-F1 difference is only 0.0066; one run is insufficient to attribute a stable advantage to augmentation.

Validation and test rankings differ: B has the highest selected validation macro-F1, whereas A has the highest observed test macro-F1. This discrepancy is consistent with an unstable evaluation under small sample support, rather than evidence that test results should be used to choose an epoch. Fig. 2 illustrates the remaining gap between training-stream and validation performance for A. Near-perfect late training-stream F1 coexists with fluctuating validation F1, indicating that the implemented perturbations do not eliminate the generalization gap.

B. Class-Resolved Outcomes

Table IV shows that aggregate gains are uneven. Both S and A correctly identify one of three buffalo treehopper observations, compared with none for B. Sampling alone predicts this class once, yielding precision 1.000 and F1 0.500. The augmented model predicts it three times, with two false positives, reducing precision and F1 to 0.333. Augmentation therefore does not improve this class relative to sampling alone.

Green leafhopper F1 increases from 0.333 to 0.500 and 0.571 under S and A, respectively. Healthy-class F1 decreases under both. Sampling alone also reduces flavescence dorée recall from 15/16 to 11/16, helping explain its lower aggregate accuracy; A recovers 14/16. These changes are compatible with a shift in class exposure and decision boundaries, but the experiment does not isolate a causal mechanism.

Crucially, every configuration has zero recall and F1 for water stress and wood diseases. In B, S, and A order, prediction counts are $(0, 1, 0)$ for water stress and $(1, 0, 1)$ for wood diseases; none of these predictions is correct. Fig. 3 shows the augmented model's confusion matrix. Its three water-stress observations are assigned to buffalo treehopper once and flavescence dorée twice, while its wood-disease observations are assigned to green leafhopper twice and healthy once. Increased exposure to rare training samples therefore does not guarantee recovery of these rare test classes.

C. Occlusion of a Misclassified Spectrum

Fig. 4 examines test row 0 using the A checkpoint. Its true label is buffalo treehopper, but the network predicts flavescence dorée with probability 0.5827. The occlusion curve contains positive and negative probability changes across the spectrum. These changes identify regions on which the incorrect decision depends under training-mean replacement. They cannot be interpreted as verified flavescence dorée biomarkers or as evidence that the predicted disease is present. The case illustrates the value of reporting the ground-truth label alongside an explanation, rather than treating a visually structured attribution as validation of the classifier.

VII. Limitations and Threats to Validity

A. Data and Statistical Validity

Only 108 observations are available for fitting a 915,014-parameter model. Each of the three focus classes has three test observations, so a single additional correct prediction changes its recall by one third. Validation contains only two water-stress observations and three wood-disease observations, making checkpoint selection sensitive to individual cases. A single seed, a single split, and inspection of 60 validation epochs do not support statistical significance claims. Independent seeds, nested validation where feasible, and a larger untouched test cohort are needed before model selection claims can be strengthened.

The class weights are a specific source of held-out-label information: they were computed from the full processed dataset. Although the same weights are used in all comparisons and no held-out spectra are used to fit standardization or augmentation statistics, this is not a fully isolated holdout protocol. A confirmatory study should recompute loss weights from training labels and repeat all configurations. A sampler-only/unweighted-loss ablation and separate noise, scaling, and offset ablations are also absent.

Exact row disjointness does not establish independence by vine, cultivar, acquisition session, or environment. The split is stratified by symptom, not grouped by these factors. Furthermore, the code averages the complete reflectance image without an explicit leaf mask, so background contributions cannot be excluded. Spatial patterns are discarded, and the concatenation in (1) introduces an artificial adjacency between two feature types. These choices restrict what can be inferred about biologically meaningful spectral structure.

B. Explanation and Deployment Boundaries

Training-mean occlusion can create spectra outside the observed distribution, and correlated neighboring bands can redistribute apparent importance. The demonstration concerns one test sample and one replacement strategy. Perturbation consistency does not establish causal interpretability or probability calibration. No inference latency, field robustness, diagnostic sensitivity in a deployment population, or prospective study has been measured.

The repository also contains an OpenCV prototype that detects a nearly horizontal diffraction band using edge detection and a Hough transform, corrects tilt, averages image intensity, and applies min–max normalization. It does not provide a verified wavelength-calibration and feature-conversion path into the 338 standardized classifier inputs. Consequently, the present results do not validate classification from diffraction photographs. An older index-based explanation script also assumes that reset test-row indices align with a separate full-dataset index table; those explanations are excluded from this study. The wavelength-occlusion path operates directly on the evaluated test features.

TABLE III
PERFORMANCE OF VALIDATION-SELECTED CHECKPOINTS ON THE FIXED TEST SET

Configuration	Best epoch	Val. macro-F1	Accuracy (%)	Balanced acc.	Macro precision	Macro recall	Macro-F1
B: weighted loss	52	0.6365	66.67	0.3785	0.3025	0.3785	0.3353
S: + sampling	27	0.5506	61.11	0.4479	0.4523	0.4479	0.4111
A: + augmentation	58	0.5941	66.67	0.4583	0.3852	0.4583	0.4177

One 60-epoch run per configuration, seed 42, $n_{\text{test}} = 36$. All configurations retain the same legacy weighted cross-entropy.

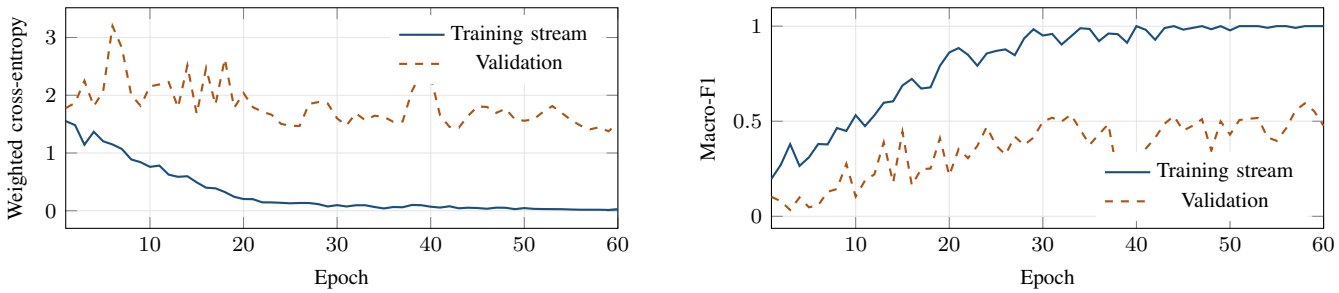


Fig. 2. Training-stream and validation history for configuration A. The epoch-58 checkpoint maximizes validation macro-F1. Training measurements include sampling and augmentation during optimization; validation samples are unchanged.

TABLE IV
PER-CLASS TEST PRECISION, RECALL, AND F1

		B: weighted loss			S: + sampling			A: + augmentation		
Class	Support	P	R	F1	P	R	F1	P	R	F1
Buffalo treehopper	3	0.000	0.000	0.000	1.000	0.333	0.500	0.333	0.333	0.333
Flavescence dorée	16	0.682	0.938	0.789	0.647	0.688	0.667	0.778	0.875	0.824
Green leafhopper	3	0.333	0.333	0.333	0.400	0.667	0.500	0.500	0.667	0.571
Healthy	8	0.800	1.000	0.889	0.667	1.000	0.800	0.700	0.875	0.778
Water stress	3	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Wood diseases	3	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

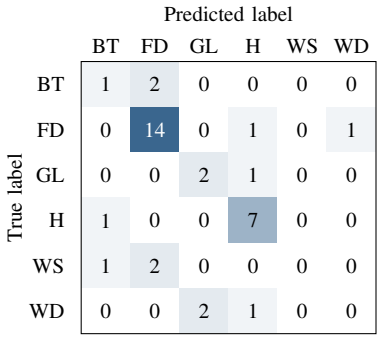


Fig. 3. Test confusion matrix for configuration A (36 observations). BT: buffalo treehopper; FD: flavescence dorée; GL: green leafhopper; H: healthy; WS: water stress; WD: wood diseases.

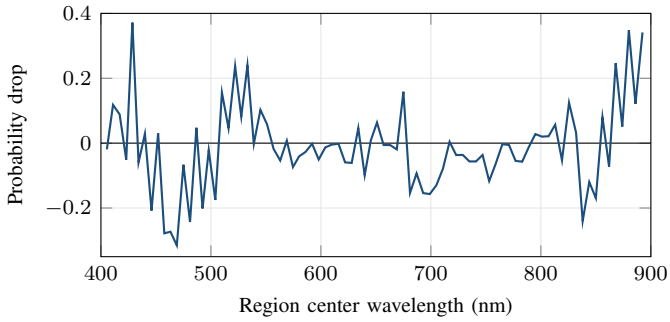


Fig. 4. Paired wavelength occlusion of test row 0, whose true label is buffalo treehopper. Configuration A incorrectly predicts flavescence dorée with probability 0.5827. Positive values indicate a reduction in that fixed predicted-class probability after replacing a five-position region with training means.

VIII. CONCLUSION

This study implemented and evaluated an imbalance-aware training and explanation workflow around an unchanged one-dimensional residual classifier for grapevine spectra. Under the existing fixed split and weighted-loss baseline, balanced sampling increased observed test macro-F1 from 0.3353 to 0.4111; adding conservative spectral augmentation yielded 0.4177 with baseline accuracy retained at 66.67%. Class-resolved analysis showed that these gains did not extend to water stress or wood diseases, both of which remained entirely unrecognized on the test set. Occlusion of a misclassified sample further demonstrated why model sensitivity should not be equated with diagnostic correctness.

The main practical result is a reproducible basis for evaluating class imbalance, not a validated field diagnosis sys-

tem. Future work should prioritize training-only loss weights, additional independent runs, grouped and external validation, reliable leaf segmentation, and focused analysis of the two unresolved classes. A calibrated connection between the image-extraction prototype and spectral classifier would require separate acquisition and validation experiments.

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